



Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

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Project Phase II

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B.Tech CSE — Initial Project Idea Submission

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Guide Details

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1. Project Title

QLoRA-Assisted Optimization of Unified VLMs for Low-Latency Hindi Assistive VQA

2. Project Abstract

2.1 Motivation

According to the WHO (World Health Organization), there are millions of individuals in India who suffer from visual impairments, which creates a high demand for accessible assistive solutions. AI driven Visual Question Answering (VQA) has emerged as a promising tool to help such people understand their surroundings. But, a significant linguistic divide exists. The majority of 'state of the art' VQA systems are optimized for English, not for low resource languages like Hindi, which is spoken by over 43% of the Indian population. Real world development introduces unique challenges like noisy photographs suffering from severe blur, poor lighting, and off center framing. Current multilingual assistive systems attempt to bridge this gap by using an external machine translation module along with existing architectures.

However, this approach introduces severe processing delays, causing the system to be unsuitable for real time assistance. This is a critical, unmet need for an assistive VQA framework that is multilingual, robust and capable of real time inference on standard hardware

2.2 Problem Statement

Current 'state of the art' multilingual VQA systems for assistive technology rely heavily on decoupled, dual encoder architectures. While these systems can achieve the baseline accuracy, they suffer from a deployment bottleneck - the sequential processing of separate vision and text modules introduces inference latency. Further, these decoupled systems usually rely on frozen vision encoders that fail to generalize against the inherent visual noise such as severe blur, poor lighting, off center framing etc. Additionally, the reliance on external machine translation to support low resource languages further compounds the processing delay.

The core problem is the inability of current decoupled multilingual VQA frameworks to provide the real time, robust and native multilingual responses required for practical, everyday assistance.

3. Objectives

1. Transition from a decoupled dual encoder baseline to a unified, end to end Vision Language architecture for assistive VQA.
2. Implement a 4 bit quantized low rank adaptation (QLoRA) training strategy to fine tune the unified model.
3. Evaluate and reduce the inference latency of the unified architecture compared to traditional decoupled pipelines.
4. Assess the unified model's ability to process multilingual (Hindi) queries and handle noisy, unstructured real world images.

4. Dataset / Data Source (if applicable)

- **Dataset Name:** VizWiz Hin (Derived Hindi version of the VizWiz dataset)
- **Source:** The base English VizWiz dataset was collected by the VizWiz project (Univer-

sity of Texas at Austin). The Hindi textual translations (questions and answers) were independently generated by our team from this English base during Phase 1.

- **Access Status:** Public (Base English Dataset) / Team Made (Hindi Translations)
- **Link:** <https://vizwiz.org/tasks-and-datasets/vqa/>
- **Description:** This dataset contains images taken by visually impaired users, paired with our Hindi questions and answers. The images feature real world visual noise (blur, poor lighting) essential for training robust assistive models.

5. Expected Deliverables

- A fully fine tuned, multilingual Vision Language Model (VLM) weight file optimized specifically for Hindi assistive VQA.
- A comparative analysis report detailing inference latency, VRAM utilization, and accuracy metrics between the Phase 1 decoupled architecture and the Phase 2 unified architecture.
- A documented methodology and reproducible codebase for hardware constrained (24GB VRAM) large model fine tuning using QLoRA.

6. Preliminary Literature Survey

No.	Paper Title	Year	Key Contribution
1	VizWiz Grand Challenge: Answering Visual Questions from Blind People	2018	Establishes the foundational benchmark dataset for assistive VQA and formally documents the inherent visual noise, such as blur and poor lighting, present in images taken by blind users.
2	Image Captioning as an Assistive Technology: Lessons Learned from VizWiz 2020 Challenge	2021	Provides adjacent assistive vision evidence showing that real world images from blind users remain difficult for traditional decoupled models, reinforcing the need for robust visual reasoning.
3	SQ-LLaVA: Self Questioning for Large Vision Language Assistant	2024	Demonstrates the effectiveness of unified, assistant style Vision Language Models, supporting the shift from decoupled pipelines to end to end multimodal reasoning.
4	A comprehensive survey of Vision Language Models (Survey)	2025	Provides background on unified VLM architectures and their ability to process high resolution visual inputs and text within a single framework.
5	A survey of efficient fine tuning methods for Vision Language Models (Survey)	2024	Reviews parameter efficient adaptation methods such as adapters and low rank approaches, motivating the use of QLoRA under hardware constraints.

6.1 Comparison of Related Work

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Gurari et al.	2018	Statistical analysis and baseline CNN models	VizWiz	Focused on English and highlighted the challenge of visual noise, but relied on standard architectures that do not address real time latency or multilingual support.
2	Pierre D et al.	2021	Assistive image captioning on blind user images	VizWiz 2020	Relevant to assistive vision, but captioning is not the same as VQA - decoupled pipelines still limit speed and robustness.
3	Guohao S et al.	2024	Unified VLM with self questioning mechanisms	Standard visual instruction datasets	Lacks specific validation for Hindi and for noisy blind user images captured in uncontrolled environments.
4	Sufyan D et al.	2025	Theoretical review of VLM architectures	N/A (Survey)	Confirms the promise of unified VLMs, but does not address Hindi assistive deployment or constrained fine tuning in practice.
5	Jialu X et al.	2024	Review of prompt, adapter, and low rank tuning methods	N/A (Survey) Discusses efficient adaptation broadly, but does not validate	Discusses efficient adaptation broadly, but does not validate QLoRA style tuning for multilingual assistive VQA.

6.2 Novelty / Research Gap Statement

Existing literature shows a clear gap between the needs of assistive VQA and the capabilities of current models. Foundational assistive work on VizWiz established that visually impaired user images are noisy and difficult to process reliably, while later assistive vision studies reinforced that traditional decoupled pipelines struggle with real world deployment. Unified Vision Language models reduce dependence on separate vision and text components and also offer a more natural path towards end to end multimodal reasoning, but they still require adaptation for Hindi, assistive domain noise and practical GPU limits. Although efficient fine tuning models such as QLoRA make model adaptation more feasible on consumer hardware, they have not yet been fully validated for low resource language assistive VQA with noisy visually impaired user

imagery. This project aims to address that gap by combining a unified VLM with 4 bit QLoRA fine tuning to enable low latency, robust Hindi assistive VQA on limited hardware

7. Potential Publication Venues

Possible Conferences

- **IEEE INDICON:** Premier IEEE conference in India, highly suitable for showcasing region specific applied AI research addressing Hindi linguistic barriers in assistive technology.
- **AAAI:** Top tier AI conference appropriate for submitting novel architectural combinations involving efficient parameter adaptation (QLoRA) and multimodal reasoning.
- **CVPR:** Leading computer vision venue. Appropriate if the dynamic resolution handling of noisy, real world assistive images yields significant visual performance improvements over traditional encoders.

Possible Journals

- **IEEE Access:** An open access journal that strongly encourages applied AI research focusing on hardware constrained optimization, accessibility, and real world deployment.
- **Expert Systems with Applications:** Welcomes practical, novel applications of modern Vision Language Models in specialized, real world assistive scenarios.
- **Pattern Recognition:** Suitable if the unified model's ability to handle visually noisy, unstructured images from blind users demonstrates distinct pattern recognition and generalization improvements.

8. References

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- [4] Danish, S., Sadeghi-Niaraki, A., Khan, S. U., Dang, L. M., Tightiz, L., & Moon, H. (2026). A comprehensive survey of Vision–Language Models: Pretrained models, fine-tuning, prompt engineering, adapters, and benchmark datasets. *Information Fusion*, 126, 103623. <https://doi.org/10.1016/j.inffus.2025.103623>
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8. Faculty Remarks

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